Efficient Multimodal Retrieval for Scientific Documents

A. Researcher and B. Researcher, University of Somewhere

Abstract

We present a pipeline for extracting structured knowledge from scientific papers that contain both text and embedded figures. Our method combines OCR-based text extraction with vision-language captioning.

1. Introduction

Scientific papers often present key results as figures rather than prose.

2. Results

Our method improves retrieval accuracy by 12% over text-only baselines. Average latency dropped from 480 ms to 210 ms after caching.

Figure 1: Accuracy improves by 12% when captioning is combined with OCR-based extraction on three benchmark datasets.

Figure 2: Latency per page with and without the caching layer.

References

[1] Researcher et al. A made-up reference, 2025.